

Factorisation Method for the Sech-Squared Potential

Source: D. Tong, *Quantum Mechanics, Problem Sheet 1, Question 8.*

Sketch the potential

$$U(x) = -\frac{\hbar^2}{m} \operatorname{sech}^2 x. \quad (1.1)$$

Show that the time-independent Schrödinger equation for a particle in this potential can be written

$$\hat{A}^\dagger \hat{A} \chi = (\mathcal{E} + 1) \chi, \quad (1.2)$$

where $\mathcal{E} = 2mE/\hbar^2$ and

$$\hat{A} = \frac{d}{dx} + \tanh x, \quad \hat{A}^\dagger = -\frac{d}{dx} + \tanh x. \quad (1.3)$$

Show, by integrating by parts, that for any normalised wavefunction χ ,

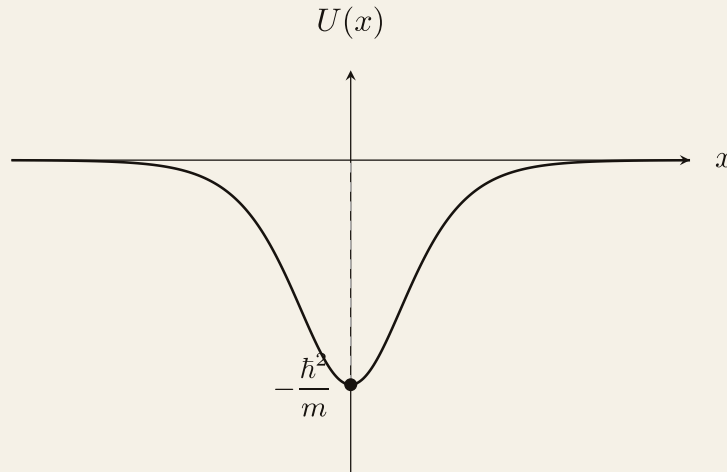
$$\int_{-\infty}^{\infty} \chi^* \hat{A}^\dagger \hat{A} \chi dx = \int_{-\infty}^{\infty} (\hat{A} \chi)^* (\hat{A} \chi) dx, \quad (1.4)$$

and deduce that the eigenvalues of $\hat{A}^\dagger \hat{A}$ are non-negative. Hence show that the ground state (with lowest energy) has $\mathcal{E} \geq -1$. Show that a wavefunction $\chi_0(x)$ is an energy eigenstate with $\mathcal{E} = -1$ iff

$$\frac{d\chi_0}{dx} + \tanh x \chi_0 = 0. \quad (1.5)$$

Find and sketch $\chi_0(x)$.

Solution.



The sech-squared potential \eqref{eq:sech-pot}

The Hamiltonian can be written as

$$\begin{aligned}
 H &= -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - \frac{\hbar^2}{m} \operatorname{sech}^2 x \\
 &= \frac{\hbar^2}{2m} \left(-\frac{d^2}{dx^2} - 2 + \tanh^2 x \right) \\
 &= \frac{\hbar^2}{2m} \left(\frac{d}{dx} + \tanh x \right) \left(-\frac{d}{dx} + \tanh x \right) - \frac{\hbar^2}{m}.
 \end{aligned} \tag{1.6}$$

Thus, we have

$$\hat{A}^\dagger \hat{A} \chi = (\mathcal{E} + 1) \chi. \tag{1.7}$$

Rather trivially following from integrating by part, we find

$$\int dx \chi^* A^\dagger A \chi = \int dx (A \chi)^* (A \chi), \tag{1.8}$$

justifying that $A^\dagger$ is indeed the Hermitian conjugate to A . Given an eigenstate of $A^\dagger A$, $|a\rangle$, with eigenvalue a . We have

$$\langle a | A^\dagger A | a \rangle = a \langle a | a \rangle = \|A|a\rangle\|^2 \geq 0 \implies a \geq 0, \tag{1.9}$$

which implies that $\mathcal{E} + 1 \geq 0$. For the ground state with energy $\mathcal{E} = -1$, we have

$$\hat{A}^\dagger \hat{A} \chi_0 = 0, \tag{1.10}$$

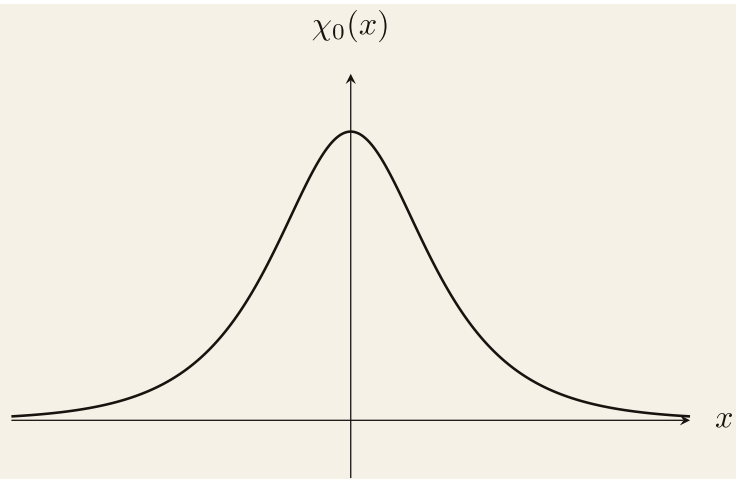
Taking the inner product of the above with χ_0 , we find

$$\langle \chi_0 | A^\dagger A | \chi_0 \rangle = \|A|\chi_0\rangle\|^2 = 0 \implies A|\chi_0\rangle = 0. \tag{1.11}$$

Solving the above, we find

$$\chi_0(x) = \frac{N}{\cosh x}, \tag{1.12}$$

where N is a normalisation constant.



Ground state wavefunction $\chi_0(x) = N \operatorname{sech} x$, peaked at the origin and decaying exponentially as $x \rightarrow \pm\infty$.